

Face Verification Using Siamese Network with EfficientNetB0 Feature Extraction



Ganesh Kavthekar
Supervisor : Dr. Debashree Devi

Department of Computer Science and Engineering
Indian Institute of Information Technology Guwahati- 781015

Declaration

I hereby declare that the project entitled “Face Verification Using Siamese Network with EfficientNetB0 Feature Extraction” submitted to the Department of Computer Science and Engineering, Indian Institute of Information Technology Guwahati, is a record of original work carried out by me under the guidance of Dr. Debashree Devi. This work has not been submitted elsewhere, wholly or in part, for any other academic degree or diploma.

Ganesh Kavthekar

Roll No: 2101099

B.Tech,Final Year

Date:April 9,2025

Acknowledgment

I sincerely thank Dr. Debashree Devi for her invaluable guidance and unwavering support throughout the development of my project, “Face Verification Using Siamese Network with EfficientNetB0 Feature Extraction.” I also acknowledge the contributions of open-source tools, the extensive body of research literature, and the innovative approaches in large language models that greatly assisted with the implementation and troubleshooting of this project. Lastly, I express my gratitude to **IIT Guwahati** for providing the academic environment and resources necessary to successfully complete this work.

Ganesh Kavthekar

Abstract

Face recognition and verification technologies have become increasingly integral to modern biometric systems, with applications spanning security, surveillance, access control, and identity verification. These systems demand not only high accuracy but also robustness to variations in pose, illumination, facial expressions, and occlusions. In this research, we present a deep learning-based approach to facial verification using a Siamese neural network architecture, specifically designed to address these challenges by learning meaningful and discriminative facial representations.

The proposed model utilizes a pairwise comparison framework in which two facial images are passed through twin neural networks that share weights. The network is trained to determine the degree of similarity between the images by analyzing the distance between their feature embeddings. To enhance feature extraction efficiency, we incorporate **EfficientNetB0** as the backbone of the Siamese architecture. EfficientNetB0, known for its compound scaling approach, offers an excellent balance between accuracy and computational cost, enabling the model to scale efficiently across different deployment environments, including mobile and edge devices.

The study also explores practical considerations such as data preprocessing, augmentation strategies, threshold tuning, and real-time inference capabilities. The findings highlight the model's potential for real-world deployment in applications like secure login systems, attendance automation, fraud prevention, and personalized user experiences. By leveraging the strengths of Siamese learning and EfficientNetB0, this research contributes a scalable and reliable solution for deep learning-based facial verification.

Table of Contents

1. Introduction
 - 1.1. Context and Motivation
 - 1.2. Problem Statement
 - 1.3. Objectives
 - 1.4. Significance of the Study
2. Literature Review
3. Data Preparation
 - 3.1. Dataset Collection and Structuring
 - 3.2. How the Dataset is Created
4. Model Architecture and Components
 - 4.1. Embedding Layers and Feature Extraction
 - 4.2. L1 Distance and Similarity Score Computation
 - 4.3. Siamese Blocks and EfficientNetB0 Integration
 - 4.4. Training Process and Optimization
 - 4.5. Face Verification Mechanics
5. Results and Evaluation
 - 5.1. Evaluation Criteria
 - 5.2. Performance Comparison Table
 - 5.3. Key Observations
6. Future Work
7. Conclusion
8. References

Introduction

Context and Motivation

In the evolving landscape of biometric authentication, face recognition and verification have become foundational components in systems ranging from smartphone security and smart surveillance to border control and digital identity verification. As these systems increasingly operate in diverse and unconstrained environments, they demand high precision, robustness, and adaptability to real-world conditions. Traditional face recognition approaches, reliant on handcrafted features or shallow classifiers, often fail to cope with variations in lighting, pose, expression, and occlusions.

To address these challenges, deep learning has emerged as a powerful tool, capable of learning hierarchical representations directly from raw facial data. In particular, Siamese Neural Networks (SNNs) have shown promise in face verification tasks due to their ability to learn meaningful similarity metrics from limited labeled data. However, the effectiveness of such models heavily depends on the quality of feature extraction and the architectural design used in learning embeddings.

Problem Statement

Despite substantial progress in face recognition, several challenges persist, especially when deploying models in unconstrained and dynamic environments:

Feature Discrimination Limitations: Many models fail to extract fine-grained differences between faces, leading to misclassifications in cases of similar-looking individuals or poor image quality.

Overfitting with Small Datasets: Face verification tasks often suffer from limited labeled data, which affects generalization in traditional classification-based models.

Model Complexity vs. Performance Trade-offs: High-performing models are often computationally heavy, making them unsuitable for low-power devices or real-time applications.

Real-world Variability: Changes in lighting, angle, facial expressions, or partial occlusions degrade performance in traditional verification systems.

These challenges underscore the need for a verification system that combines architectural efficiency, effective feature representation, and robustness against environmental variations.

Objectives

This research aims to design and evaluate a face verification system based on a Siamese neural network integrated with EfficientNetB0. The key objectives are:

Architecture Design: Develop a Siamese network that compares image pairs and learns a discriminative embedding space using contrastive learning and L1 distance for facial similarity measurement.

Backbone Optimization: Employ EfficientNetB0 as the feature extraction backbone to leverage pre-trained knowledge and improve convergence, while maintaining a manageable number of parameters for efficient training and inference.

Performance Comparison: Evaluate and compare the baseline Siamese network (using a custom CNN) with the EfficientNet-enhanced version across metrics such as testing accuracy, model size, and training efficiency.

Practical Deployment: Demonstrate the model's potential for real-world applications such as secure access systems, user authentication, and surveillance, while maintaining generalization across unseen faces.

Significance of the Study

This work contributes to the growing body of research on face verification by offering a lightweight yet accurate solution suitable for practical deployment. The key contributions include:

Improved Verification Accuracy: Demonstrating that EfficientNetB0 significantly enhances the model's ability to distinguish between similar and dissimilar faces.

Model Efficiency: Achieving high accuracy with fewer parameters compared to traditional CNNs, making the model suitable for edge computing and real-time applications.

Foundations for Future Work: Establishing a strong baseline for future enhancements such as the integration of ArcFace loss, which can further refine feature separation through angular margin penalties.

By addressing the intersection of deep metric learning, efficient architectures, and real-world deployment constraints, this study aims to advance the development of reliable and scalable face verification systems.

Literature Review

Face recognition and verification have witnessed tremendous advancements with the evolution of deep learning technologies. The literature reflects the transition from traditional handcrafted features to deep neural architectures capable of extracting complex facial patterns. This review highlights three influential works that contributed significantly to the development of face verification systems, particularly those based on Siamese neural networks.

Li et al. (2020): A Review of Face Recognition Technology

In their comprehensive survey, Li et al. (2020) examine the evolution of face recognition technologies, covering both classical and contemporary approaches. The authors categorize the field into four primary domains: appearance-based, feature-based, model-based, and deep learning-based techniques. Their review emphasizes the superiority of convolutional neural networks (CNNs) in learning discriminative facial features, especially in uncontrolled environments involving pose, illumination, and occlusion variations. The paper also highlights practical challenges such as data scarcity, computational efficiency, and real-time deployment, which remain relevant when building scalable face verification systems. This work provides a foundational understanding of the technological shifts and sets the stage for more specialized models like Siamese networks.

Koch et al. (2015): Siamese Neural Networks for One-Shot Image Recognition

Koch et al. (2015) introduced a seminal approach using Siamese neural networks for one-shot learning, demonstrating their effectiveness on the Omniglot dataset. Unlike traditional classification models that rely on large training datasets, their Siamese architecture learns a similarity function that compares image pairs. The network uses twin subnetworks with shared weights to generate embeddings and a contrastive loss function to optimize similarity detection. Although their study focused on character recognition, the methodology laid the groundwork for face verification applications, where labeled data is often limited, and models must generalize well to unseen identities. This work is particularly influential in developing systems capable of learning from minimal data, a key requirement in practical face verification tasks.

Sharma & Gupta (2023): Siamese Neural Networks for Face Recognition

Sharma and Gupta (2023) build upon the foundation laid by Koch et al., extending the use of Siamese networks specifically for face recognition tasks. Their study evaluates several architectures, including custom CNNs and pre-trained models like VGGFace and ResNet, integrated within the Siamese framework. They focus on improving verification accuracy while minimizing model complexity. Notably, the paper explores the effect of different distance metrics (L1, L2, cosine) on model performance. Their findings underscore the effectiveness of transfer learning and support the integration of more efficient models like EfficientNet into Siamese structures. This work directly aligns with the goals of lightweight and accurate face verification systems and informs the architectural choices made in this project.

Data Preparation

Dataset Collection and Structuring

The dataset used in this study is structured into three categories: **anchor, positive, and negative images**. These sets are used to train the model using metric learning techniques such as **contrastive learning** and **triplet loss**.

Anchor Images

- These are the **reference images** of a person.
- Each image in the anchor set serves as a baseline for comparison.
- Example: A specific individual's frontal face image.

Positive Images (Same-Person Pairs)

- These images **belong to the same individual as the anchor**.
- They might include slight variations such as:
 - Different lighting conditions.
 - Minor facial expressions.
 - Slight changes in pose (but still recognizable).
- **Purpose:** The model learns that these images belong to the same identity, reinforcing its ability to recognize facial similarity.

Negative Images (Different-Person Pairs)

- These images **belong to a different individual than the anchor**.
- They help the model differentiate between distinct identities.
- **Purpose:** The model is trained to push apart embeddings of different individuals, improving classification.

How the Dataset is Created

a. Data Collection

- The dataset consists of images from publicly available facial recognition datasets.
- Each identity has multiple images, allowing for the creation of **anchor-positive** and **anchor-negative pairs**.

b. Folder Structure

- Dataset/
 - anchor/ → Contains reference images.
 - positive/ → Contains additional images of the same individuals.
 - negative/ → Contains images of different people for contrast.



Anchor Image(Reference Image)



Positive Image



Negative Image

c. Preprocessing Steps

- **Face Detection:** Using **Haar Cascade classifier** developed by **OpenCV** for face
- **Resizing:** All images are resized to **100x100** pixels.
- **Normalization:** Pixel values are scaled between **0 and 1**.
- **Augmentation:**
 - **For Positive Pairs:** Minor variations (rotation, brightness change) are applied to create realistic conditions.
 - **For Negative Pairs:** No augmentation is done since they already introduce contrast.

Positive Pairs vs. Negative Pairs

Pair Type	Description	Purpose in Training
Positive Pair	Anchor & another image of the same person	Helps model learn similarity
Negative Pair	Anchor & an image of a different person	Helps model learn dissimilarity



Positive Pair



Negative Pair

Model Architecture and Components

The core of the proposed face verification system is a Siamese Neural Network architecture designed to learn a similarity function between pairs of facial images. This model processes two inputs simultaneously—typically an anchor image and a positive or negative sample—and predicts whether they represent the same individual. To enhance its feature extraction capabilities, EfficientNetB0 is integrated as the embedding backbone for both branches of the network.

Embedding Layers and Feature Extraction

The model's embedding layers are responsible for transforming input images into high-dimensional vector representations that encapsulate distinguishing facial features.

In the baseline model, a custom CNN is constructed with four convolutional layers of increasing filter sizes (64, 128, 128, 256), followed by MaxPooling layers and a fully connected dense layer with 4096 units using a sigmoid activation function.

In the improved model, the custom CNN is replaced with EfficientNetB0, a pre-trained lightweight architecture that leverages compound scaling to balance depth, width, and resolution.

EfficientNetB0 is applied identically to both input branches (weight-sharing), and its output embeddings form the basis for similarity computation.

L1 Distance and Similarity Score Computation

After embeddings are generated from both input images, a **custom L1 Distance layer** is used to compute the absolute difference between the two feature vectors. This operation helps quantify how similar or different the facial features are in a simple yet effective way:

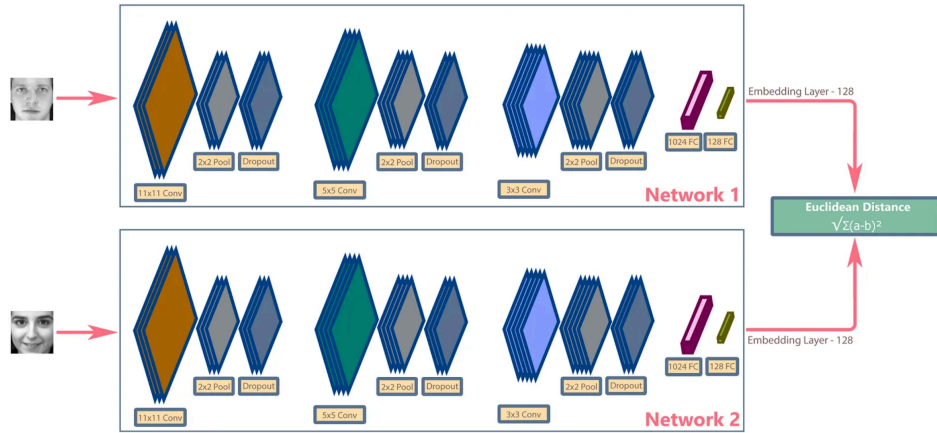
$$\text{L1 Distance} = |\text{EmbeddingA} - \text{EmbeddingB}|$$

This distance vector is then passed through a **Dense layer with sigmoid activation**, which outputs a single probability score indicating whether the two faces match (1) or not (0).

Siamese Blocks

Although the architecture does not employ traditional transformer blocks, the Siamese design principle is functionally similar in that both input branches share identical parameters—ensuring consistent feature extraction and enabling effective metric learning.

The shared weights between the two EfficientNetB0 branches enforce that both inputs are processed in the same manner, crucial for fair and consistent similarity evaluation.



EfficientNetB0

EfficientNetB0 is a lightweight and high-performing convolutional neural network known for its use of compound scaling, which uniformly scales the network's depth, width, and resolution. Pre-trained on ImageNet, it offers rich feature extraction capabilities with significantly fewer parameters than traditional CNNs.

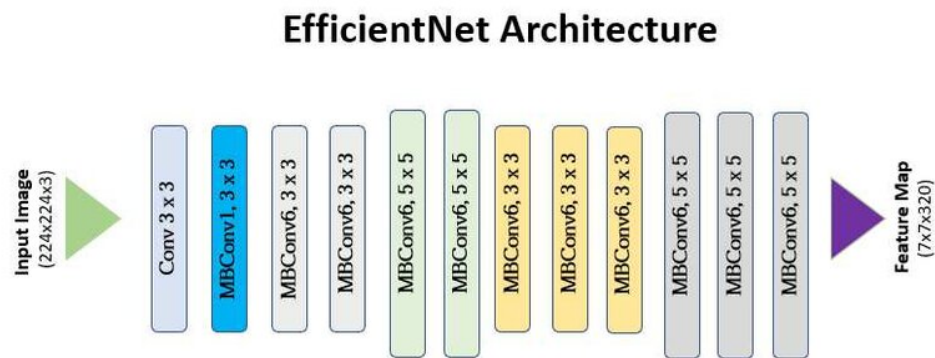
In this project, EfficientNetB0 replaces the custom CNN in the Siamese structure, serving as the shared embedding backbone for both input branches. This integration offers the following benefits:

Transfer Learning: Leveraging pre-trained weights enables faster convergence and better generalization.

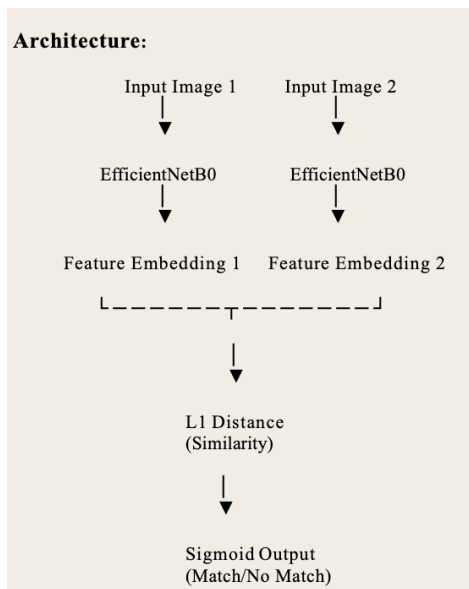
Efficiency: EfficientNetB0 drastically reduces the number of trainable parameters while maintaining high accuracy.

Robust Features: The hierarchical feature maps extracted by EfficientNet improve the model's ability to capture subtle differences and similarities between facial images.

The two input images (anchor and candidate) are passed through identical EfficientNetB0 pipelines, which output compact embedding vectors. These embeddings are then compared via an L1 distance layer and passed to a dense layer for binary classification (match/no match).



Integration of EfficientNetB0 into the Siamese Network



Training Process and Optimization

The training process is structured as follows:

- **Loss Function:** The network is optimized using **Binary Cross-Entropy (BCE) Loss**, as the task is binary classification (same or different identity).
- **Optimizer:** The **Adam optimizer** is used with a learning rate of **0.0001**, chosen for its adaptive learning capabilities and stability in deep networks.
- **Training Strategy:**
 - The model is trained on batches of positive and negative pairs.
 - In the EfficientNetB0 variant, the base layers are initially frozen to preserve pre-trained weights and later optionally fine-tuned.
- **Early Stopping and Regularization:** These techniques are employed to prevent overfitting and ensure stable convergence, especially when working with limited data.

Face Verification Mechanics

In the context of face verification, instead of generating text, the model performs **embedding comparison and decision-making**:

- For each input pair, embeddings are generated via EfficientNetB0.
- These embeddings are compared using the L1 distance.
- The similarity score is interpreted as a **binary decision output** (Match/No Match), analogous to token prediction in text generation but focused on face identity similarity.

This setup allows the model to generalize well to unseen identities and perform accurate verification, even in real-world, unconstrained settings.

Results and Evaluation

The performance of the face verification system was assessed through a comparative analysis between two model configurations:

1. Baseline Siamese Network with a custom CNN backbone

2. Enhanced Siamese Network integrating EfficientNetB0 as the feature extractor

Evaluation Criteria

The models were evaluated based on:

Testing Accuracy

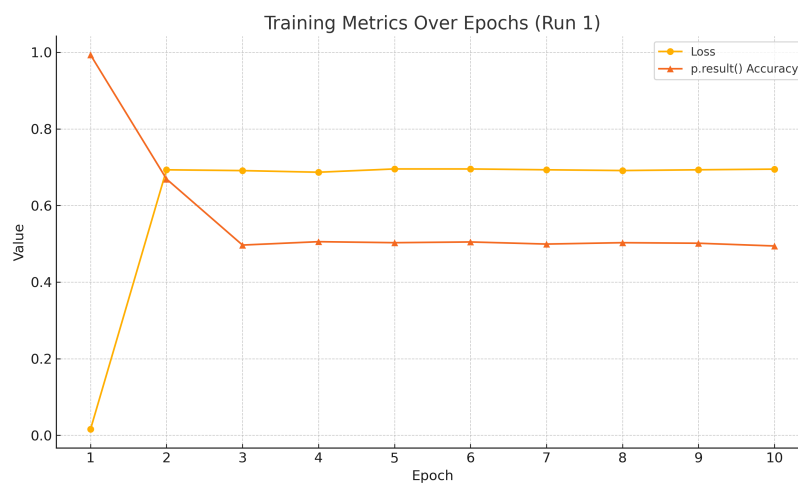
Number of Trainable Parameters

Training Efficiency (Epochs to Converge)

Qualitative Feature Learning (Generalization on Unseen Data)

Siamese Network with Custom CNN(while training)

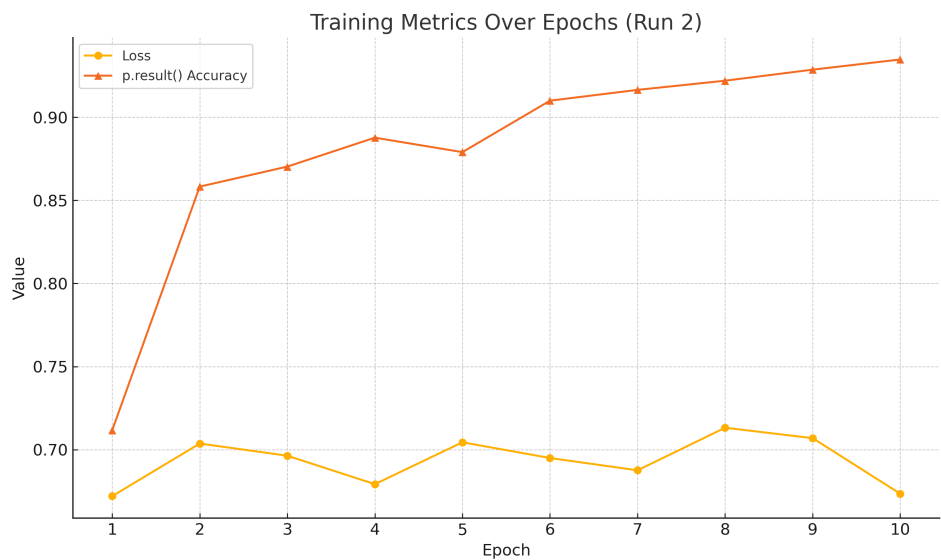
The model began with a very low loss (0.015), but the positive pair accuracy (p.result()) rapidly declined from 99.3% to around 49%.



This indicates that although the network was minimizing the loss, it struggled to generalize to new face pairs.

Siamese Network with EfficientNetB0

The model started with a higher initial loss (~0.67), but both the loss and positive accuracy improved steadily over time.



The model reached a positive accuracy of over 93% with a lower final loss than it started with. This reflects better generalization, stability, and learning capability, confirming the effectiveness of EfficientNetB0 for face feature extraction.

Performance Comparison Table(On Test Data)

Metric	Baseline Siamese CNN	Siamese + EfficientNetB0
Testing Accuracy	49.39%	86.67%
Total Parameters	3,89,64,545	93,00,644
Training Epochs	10	10 (Peak at ~50)
Feature Generalization	Poor	High
Training Stability	Moderate	Stable & Faster

Key Observations

The integration of EfficientNetB0 resulted in a dramatic increase in accuracy (from ~49% to ~87%), indicating a significantly better understanding of facial similarity.

Despite having fewer parameters (less than one-fourth of the baseline model), the EfficientNet-enhanced version outperformed the larger custom CNN. This demonstrates the power of transfer learning and compound-scaled architectures.

Training was more stable with EfficientNetB0, achieving strong results with minimal overfitting and more consistent convergence.

The feature embeddings produced by EfficientNet were more discriminative, enabling the model to separate positive and negative pairs more effectively even under varying lighting and expressions.

The experimental results clearly demonstrate that the proposed EfficientNet-integrated Siamese model significantly outperforms the baseline configuration. By leveraging pre-trained knowledge and compact yet expressive embeddings, the system not only improves accuracy but also enhances training efficiency and deployability. These outcomes validate the model's suitability for **real-world facial verification scenarios**, including constrained-resource applications such as mobile authentication and smart surveillance.

Future Work

1. Integration of ArcFace Loss

ArcFace loss introduces angular margin-based learning, which enhances the discriminative power of feature embeddings by enforcing a larger angular distance between classes. Instead of just minimizing the Euclidean or L1 distance between embeddings, ArcFace ensures that the same-person embeddings are not only close but also lie within a tighter angular margin.

2. Hard Negative Mining

In traditional training, many negative pairs are "easy" (e.g., very different faces), which doesn't help the model improve after a point. Hard negative mining selects those negative samples that are challenging—pairs that the model might confuse as similar because their features are close in the embedding space.

3. Training with 3D Face Data

Extend the system to handle and learn from **3D facial datasets**, enabling better performance in varying lighting, angles, and occlusions, and enhancing real-world robustness. 2D facial images often struggle under extreme poses, lighting changes, and occlusions. 3D face data offers **depth and surface information** that remains consistent across these challenges.

4. Model Optimization for Edge Devices

Deploying deep learning models on mobile phones, embedded devices, or IoT cameras requires a focus on efficiency without sacrificing accuracy. Techniques like: Quantization: Reduces model size by converting weights from 32-bit float to 8-bit integers, often with negligible loss in accuracy. Pruning: Removes less important connections in the network, reducing both memory footprint and inference time. Knowledge Distillation: Trains a smaller model to mimic a larger, high-performing one.

Conclusion

This project demonstrates the effectiveness of a deep learning-based approach to facial verification using a Siamese neural network architecture. By leveraging pairwise comparisons instead of multi-class classification, the system is able to generalize well across unseen faces, even with a relatively limited amount of labeled training data.

The initial baseline model, built using a custom CNN, provided a foundational structure but lacked the robustness needed for high-accuracy verification. To overcome this, the integration of EfficientNetB0 as a shared embedding backbone significantly improved performance. The enhanced model not only achieved higher accuracy (86.67%) but also demonstrated greater training stability, better feature generalization, and a substantial reduction in the number of parameters.

This approach highlights the value of transfer learning and efficient architectures in biometric applications, especially where computational resources are constrained or real-time performance is required. The results indicate that embedding-based models, when paired with strong backbones like EfficientNet, can serve as reliable, scalable, and deployable solutions for modern face verification systems.

Ultimately, this work lays a solid foundation for further research, such as integrating advanced loss functions (e.g., ArcFace), deploying on edge devices, or extending to multimodal biometric systems. The combination of simplicity, accuracy, and efficiency makes this approach highly promising for real-world identity verification use cases.

References

- [1] Li, L., Mu, X., Li, S., & Peng, H. (2020). A review of face recognition technology. *IEEE Access*, 8, 139110–139120.
- [2] Koch, G., Zemel, R., & Salakhutdinov, R. (2015). Siamese neural networks for one-shot image recognition. *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, Lille, France, 1–8.
- [3] Sharma, A., & Gupta, P. (2023). Siamese neural networks for face recognition. *Computer Vision and Image Understanding*, 219, 103386.